

biometric sensors, printed sensors, and MEMS and NEMS sensors.

IoT sensors. IoT sensors include temperature sensors, proximity sensors, pressure sensors, RF sensors, pyroelectric infrared (PIR) sensors, water-quality sensors, chemical sensors, smoke sensors, gas sensors, liquid-level sensors, automobile sensors and medical sensors. These sensors are connected to a computer network for monitoring and control purposes.



Fig. 2: PM2.5/PM10 sensor (Courtesy: <http://aqicn.org>)



Fig. 3: PM2.5 sensor (Courtesy: www.digikay.com)



Fig. 4: Grain-size RFID chip (Courtesy: <https://voiceofpeopletoday.com>)

Using sensors and the Internet, IoT systems have wide applications across industries with their unique flexibility in providing enhanced data collection, automation and operation.

The global market for IoT sensors reached \$7.3 billion in 2015. It is expected to reach \$47.8 billion in 2021 from nearly \$10.6 billion in 2016, growing annually at 35 per cent during 2016-21. The Asia-Pacific market for IoT sensors is expected to grow from \$3 billion in 2016 to \$14 billion in 2021 at a CAGR of 36.1 per cent from 2016 through 2021.

Pollution sensors. Air pollution sensors are used to detect and monitor the presence of air pollution in the surrounding area. These can be used for both indoor and outdoor environments. Although there are various types of air pollution sensors, most of these sensors focus on five parameters: particulate matter, ozone, carbon monoxide, sulphur dioxide and nitrous oxide. These sensors are somewhat costly but are becoming more affordable for common use.

Sensors capable of detecting particulate matter with a diameter between 2.5 and 10µm (PM10) and a diameter less than 2.5µm (PM2.5) are available in the market. Fig. 2 shows a typical PM sensor popular among hobbyists and experimenters. Fig. 3 shows an easy-to-assemble PM2.5 sensor from Sharp Corporation with a detection time of ten seconds.

RFID sensors. RFID chips (Fig. 4) as small as the size of rice grains can be inserted directly under the skin for

use as ID cards. There is a trend to use RFID chips in many products including contactless banks cards and Oyster cards. There are also cases where chips are implanted in pets and cattle for monitoring.

Wearable sensors. These sensors include medical sensors, GPS, inertial measurement unit (IMU) and optical sensors. With modern techniques and miniature circuits, wearable sensors can now be deployed in digital health monitoring systems. Sensors are also integrated into various accessories such as cloths, wrist bands, eyeglasses, headphones and smartphones. An IDTechEx report forecasts optical, IMU and GPS sensors to dominate the sensors market in terms of revenue by 2022 (Fig. 5).

Wearable applications and IoTs are expected to drive double-digit growth in the global market for sensors. Due to decrease in manufacturing costs and low power consumption of sensors, most traditional wired connections will be replaced with wireless sensors and integrated into wireless networks in the future.

Image sensors. The best example of this sensor is found in your smartphone camera. An image sensor detects and conveys the information that constitutes an image. Digital imaging is fast replacing analogue imaging. Most digital cameras use CMOS sensors, which allow faster speed with lower power consumption.

An image sensor from Renesas is shown in Fig. 6.

Biometric sensors. The most common biometric sensor is your fingerprint module. R30x fingerprint module is quite popular among hobbyists and experimenters. The latest generation of fingerprint sensors from Qualcomm consists of sensors for display, glass and metal, detection of directional gestures, and underwater fingerprint match and device wake-up. These sensors are designed as both an integrated solution with Qualcomm Snapdragon mobile platforms, and as standalone sensors that can be used with other non-Snapdragon platforms. Qualcomm fingerprint sensors for glass and metal are likely to reach the market in the first half of 2018.

Printed sensors. Sensors printed on flexible substrates are becoming popu-

